清華大學深圳国际研究生院
Tsinghua Shenzhen International Graduate School

清華大學
Tsinghua University

Programming Challenge: blind Projector from 6D Space to 2D Pixels

SIGS：Self-learned Implicit Geometric System

姓名：王孟哲　　王振羽

日期：6 / 7 / 2025

清華大學深圳国际研究生院
Tsinghua Shenzhen International Graduate School

Leaderboard

Blind Projector

Showing results for: **Blind Projector**

Note: For this problem, a lower score is better.

Rank	User	Score
1	wmz_wzy	30.7384
2	xhr_zl	30.7406
3	zy_wxy	30.9568
4	lrx_czy	31.0388
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8	cjd_yth	31.1878
9	zhy-jty	31.5017
10	zjy_xyf	31.6046
11	gzr_qzh	31.7282
12	zqr_cqx	32.3640

We are here!

先说结果： We Are SOTA !!!

Code：https://github.com/WMZ0518/Embodied-AI

目录

一、题目分析

二、实现细节

三、实验结果

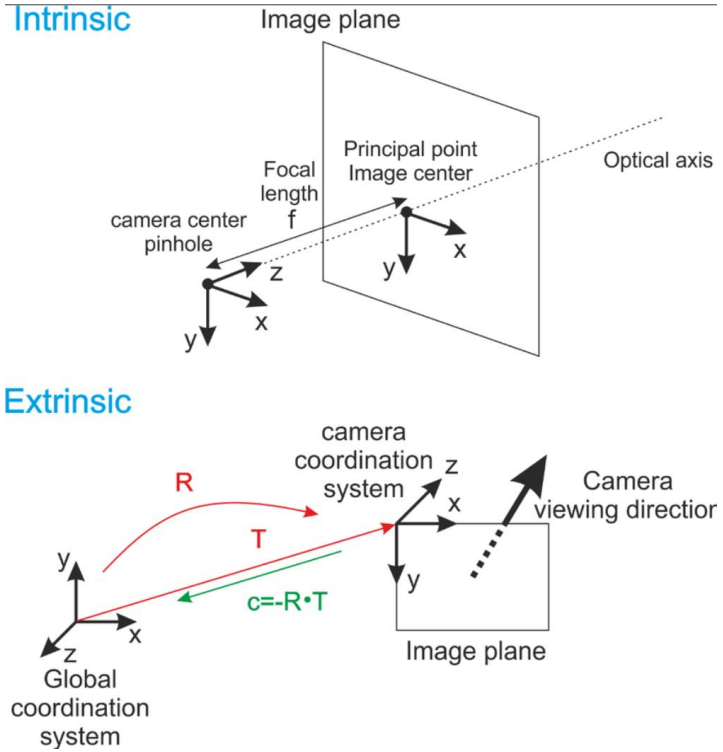
四、讨论分析

第一部分

题目分析

题目理解

Camera_id这一特征重要吗？为什么？



相机内参和外参影响成像

设世界坐标点为$\mathbf{X_W} = [X,Y,Z,1]^T$,相机投影矩阵为$\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$，则对应的齐次像素坐标$\mathbf{x} = [u,v,1]^T$满足：

$$\mathbf{x} \sim \mathbf{P}\,\mathbf{X}_w = \mathbf{K}\,[\mathbf{R}|\mathbf{t}]\,\mathbf{X}_w$$

当展开为非齐次坐标时：

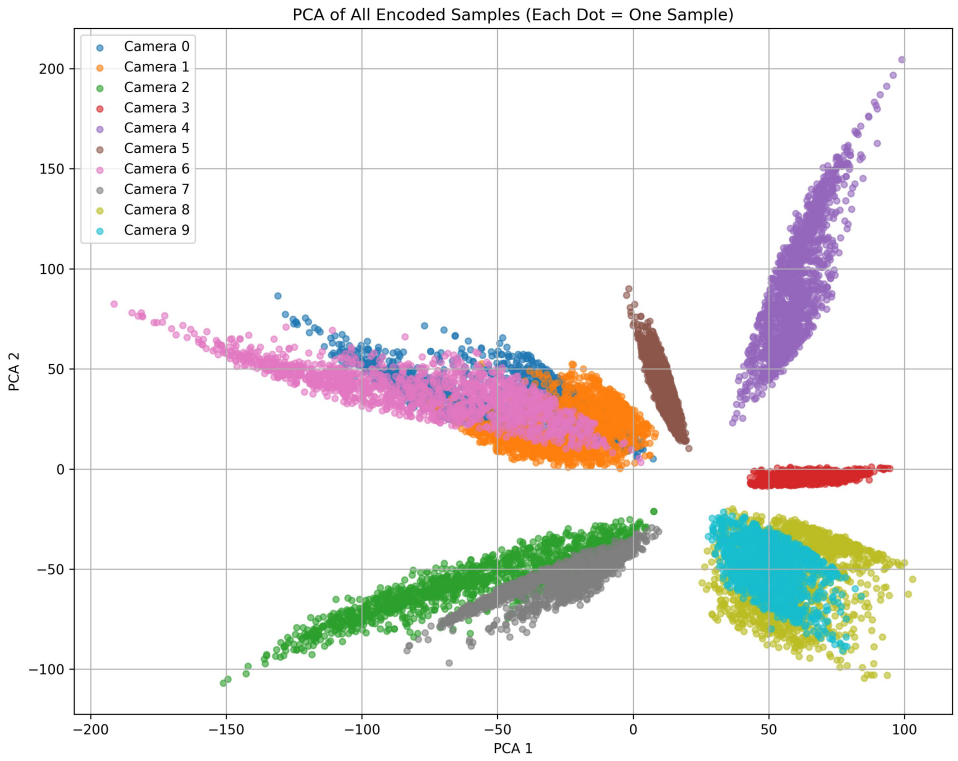
$$\begin{bmatrix} u \\ v \end{bmatrix} = \frac{1}{Z_c}\begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \end{bmatrix}\left(\mathbf{R}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \mathbf{t}\right)$$

其中**R**，**t**为相机中的外参
K是相机的内参矩阵，包括焦距、主点、斜切等参数

在物理层面来讲，相机的内参和外参对最终的投影坐标是有影响的

题目理解

Camera_id这一特征重要吗？为什么？



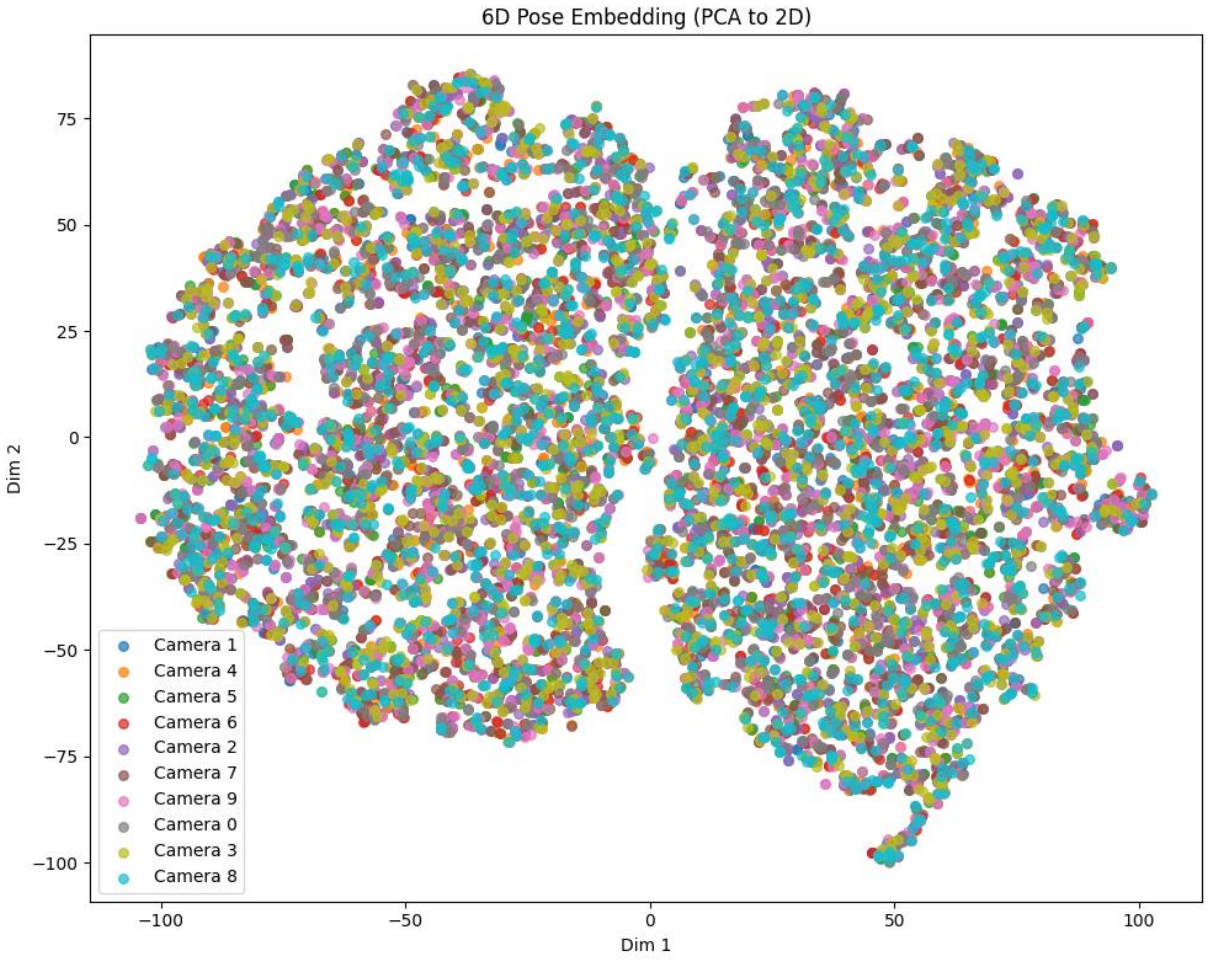
在数据层面来讲，相机的内参和外参对最终的投影坐标是还是被证明有影响的。但本次题目的训练集提供了camera_id，但是测试集没有提供camera_id，因此，我们需要找到一种大一统的方法，这个方法要尽量适用于所有camera_id的信息，又要尽量保留camera_id的信息。

题目理解

为什么不直接去预测Camera_id？

在物理层面来说，用什么相机和在物理世界中的三维空间坐标是没有任何关系的。

在数据层面来说，我们对6维坐标进行了降维，降至2维后，进行可视化，我们发现实际上是没有关系的。更加印证我们任务坐标与相机没有关系的猜想。



题目理解——如何对欧拉角建模？

2.5.4 $z-y-x$欧拉角

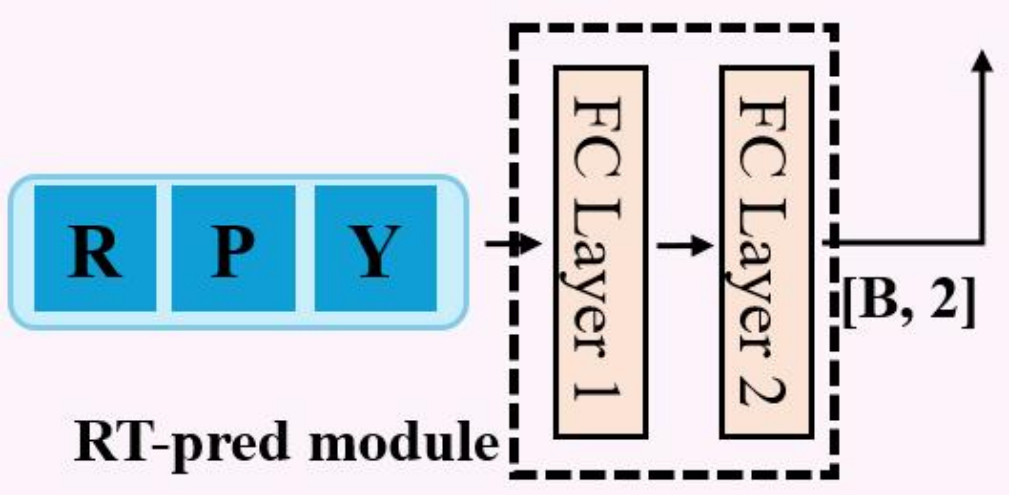
每次转动都是相对于**运动坐标系的某轴**进行的，**而不是固定的参考系{A}**，这样的三次转动称为**欧拉角**，又因转动的顺序是绕z轴、y轴、x轴，故这样的描述法称为**z-y-x欧拉角**。由于三次旋转都是相对运动坐标系进行的，根据“从左向右(右乘)”的原则，则得到相应的旋转矩阵为：

$$ {}^{A}_{B}R_{ZYX}(\alpha,\beta,\gamma) = R(Z,\alpha)R(Y,\beta)R(X,\gamma) $$

$$ {}^{A}_{B}R_{ZYX}(\alpha,\beta,\gamma) = \begin{bmatrix} cos\alpha & -sin\alpha & 0 \\ sin\alpha & cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos\beta & 0 & sin\beta \\ 0 & 1 & 0 \\ -sin\beta & 0 & cos\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\gamma & -sin\gamma \\ 0 & sin\gamma & cos\gamma \end{bmatrix} $$

$$ {}^{A}_{B}R_{ZYX}(\alpha,\beta,\gamma) = \begin{bmatrix} c\alpha\cdot c\beta & c\alpha\cdot s\beta\cdot s\gamma - s\alpha\cdot c\gamma & c\alpha\cdot s\beta\cdot c\gamma + s\alpha\cdot s\gamma \\ s\alpha\cdot c\beta & s\alpha\cdot s\beta\cdot s\gamma + c\alpha\cdot c\gamma & s\alpha\cdot s\beta\cdot c\gamma - c\alpha\cdot s\gamma \\ -s\beta & c\beta\cdot s\gamma & c\beta\cdot c\gamma \end{bmatrix} $$

这一结果与绕固定轴$x-y-z$旋转的结果**完全相同**，这是因为绕固定轴旋转的顺序与绕运动轴旋转的**顺序相反**，且旋转的**角度也对应相等**，因此得到的变换矩阵是相同的。

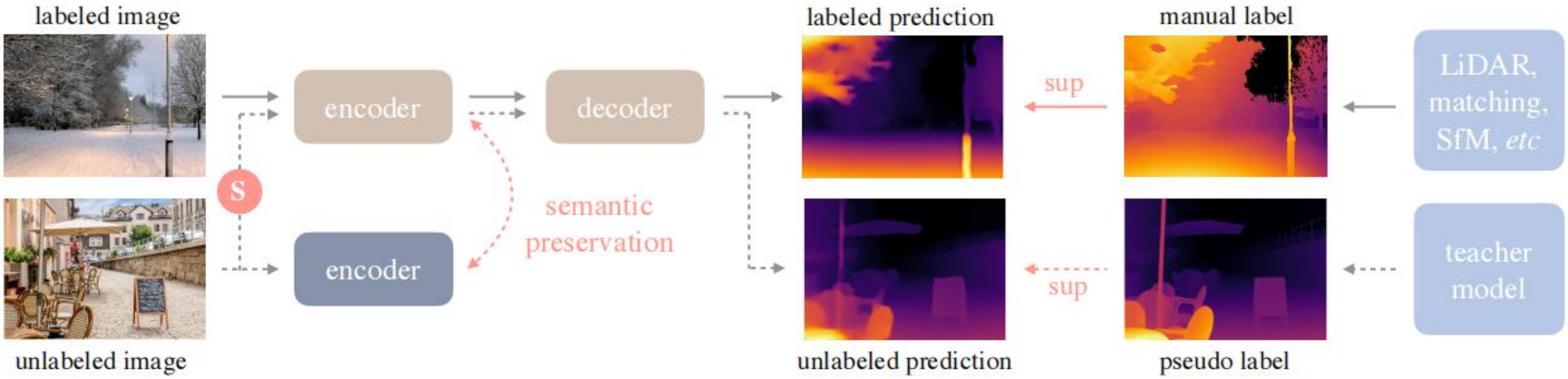


第二部分

实现细节

启发idea

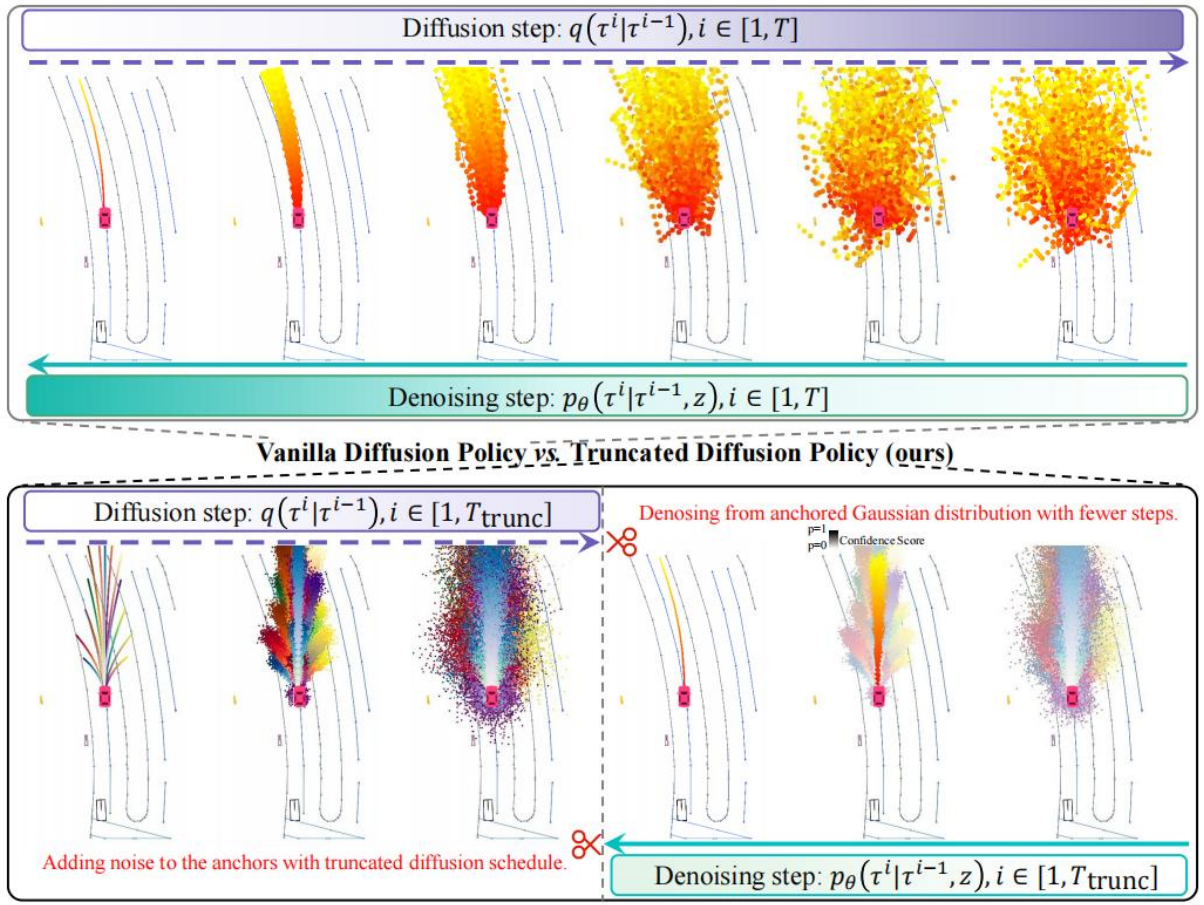
语义先验 → 深度检测？



Piccinelli, L., Yang, Y. H., Sakaridis, C., Segu, M., Li, S., Van Gool, L., & Yu, F. (2024). **UniDepth**: Universal monocular metric depth estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 10106-10116).

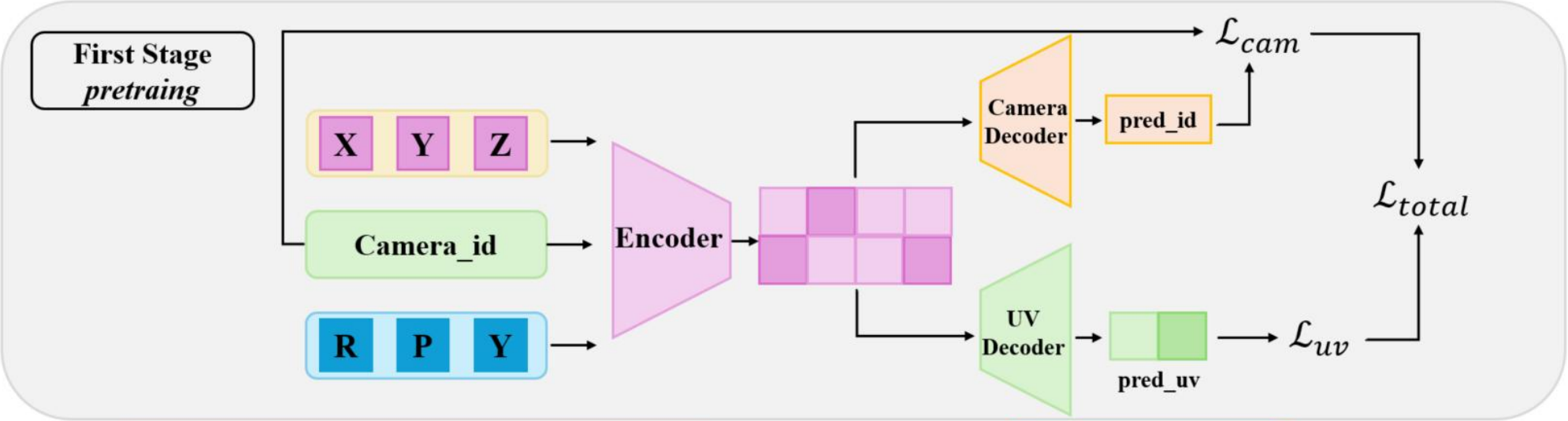
启发idea

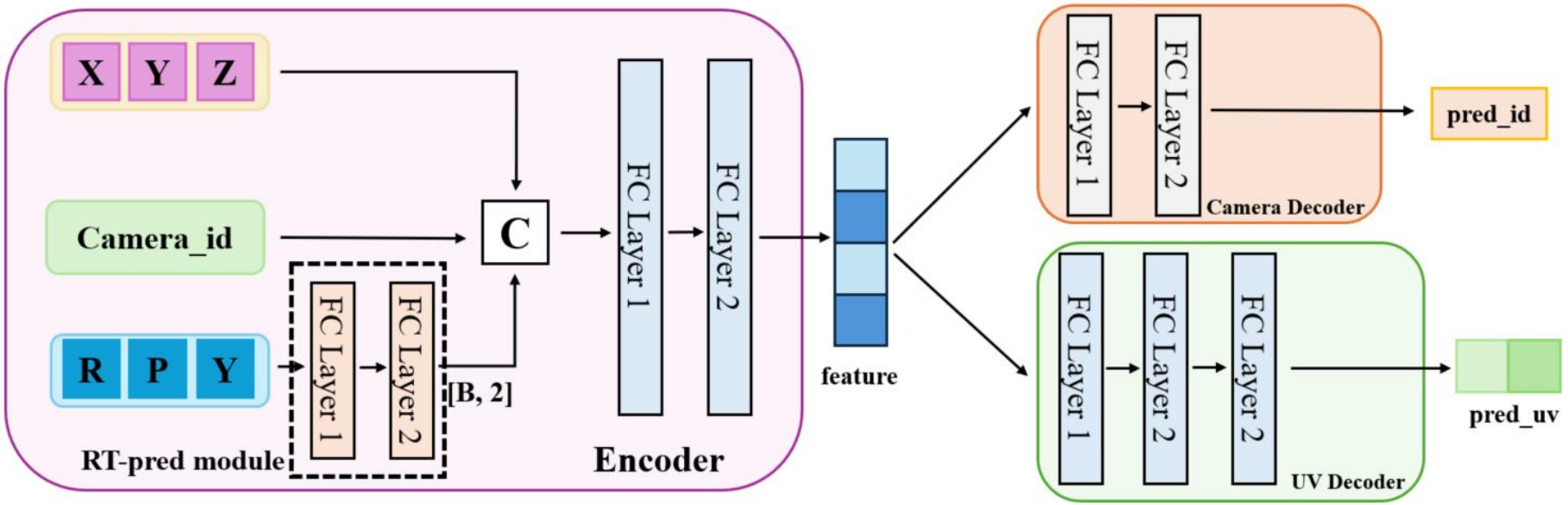
扩散锚点 → 易于收敛？



Liao, B., Chen, S., Yin, H., Jiang, B., Wang, C., Yan, S., ... & Wang, X. (2025). Diffusiondrive: Truncated diffusion model for end-to-end autonomous driving. In Proceedings of the Computer Vision and Pattern Recognition Conference (pp. 12037-12047).

模型架构——stage1

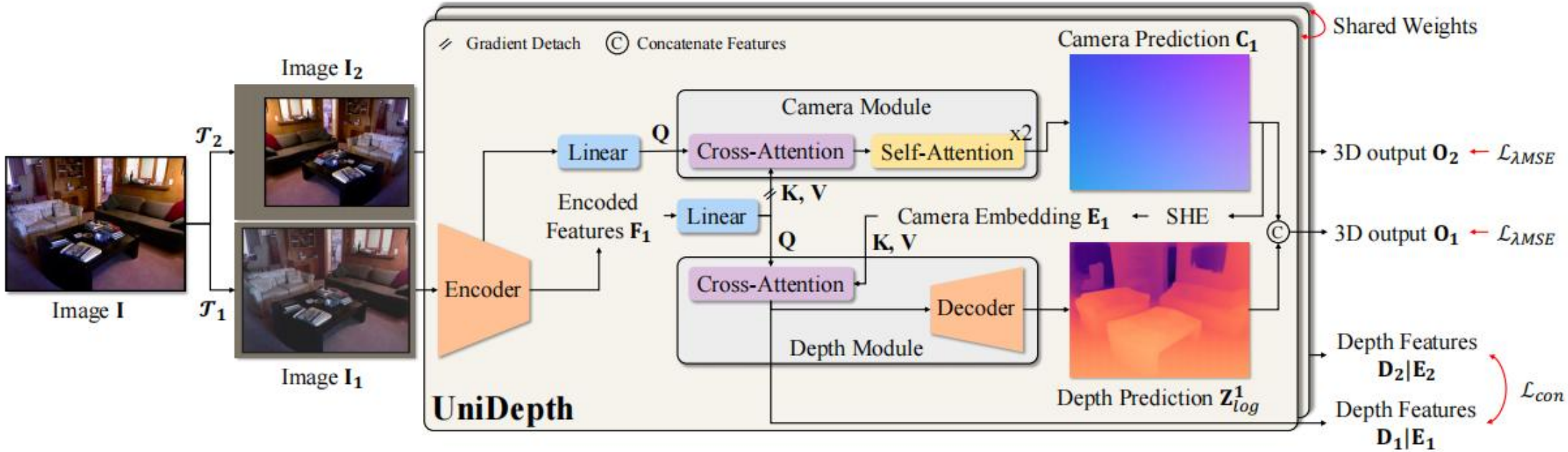

First Stage
pretraing
X
Y
Z
Camera_id
R
P
Y
Encoder
Camera Decoder
pred_id
UV Decoder
pred_uv
$\mathcal{L}_{cam}$
$\mathcal{L}_{total}$
$\mathcal{L}_{uv}$


X
Y
Z
Camera_id
R
P
Y
C
FC Layer 1
FC Layer 2
[B, 2]
RT-pred module
FC Layer 1
FC Layer 2
Encoder
feature
FC Layer 1
FC Layer 2
Camera Decoder
pred_id
FC Layer 1
FC Layer 2
FC Layer 2
UV Decoder
pred_uv

如何让 model“忽视”camera_id
找到一个通用的“投影表征”

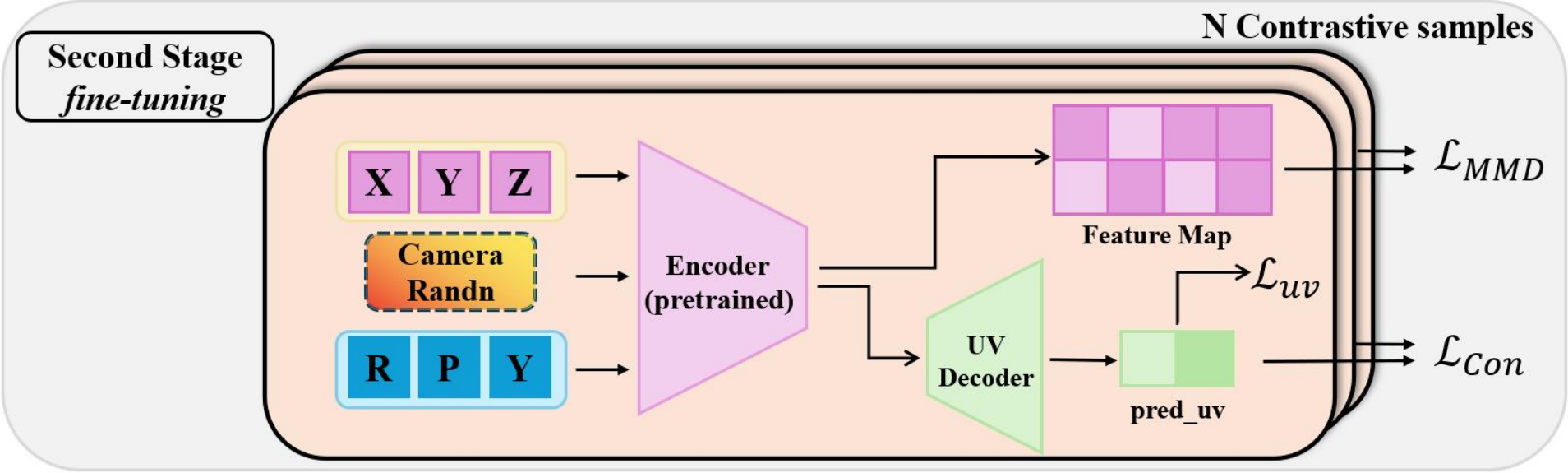
?

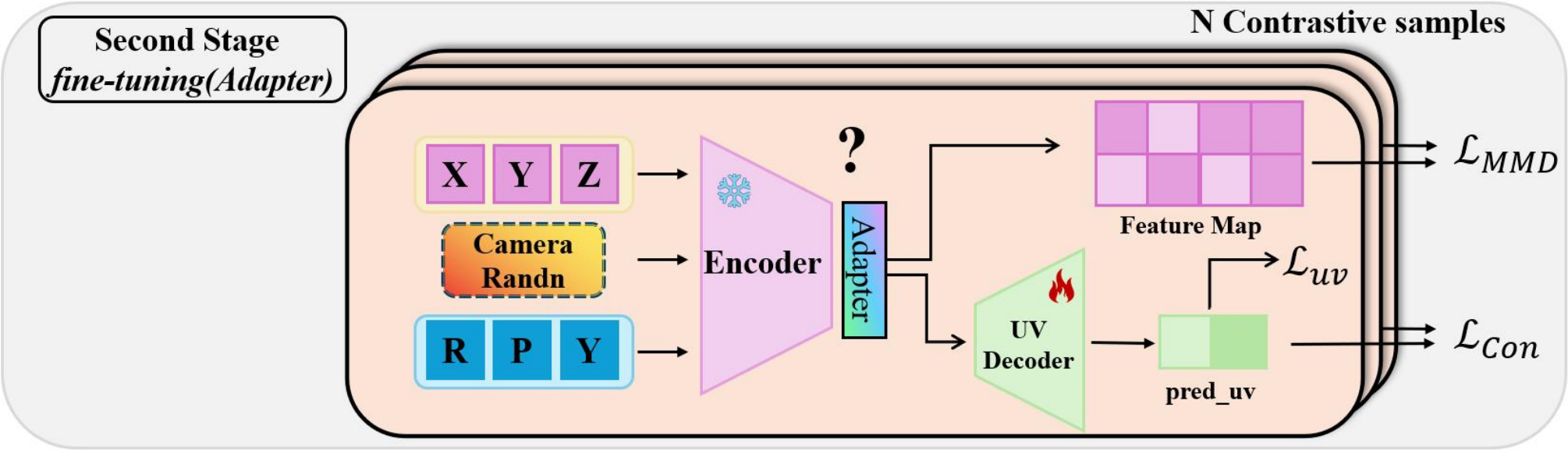
启发idea



Yang, L., Kang, B., Huang, Z., Xu, X., Feng, J., & Zhao, H. (2024). **Depth anything:** Unleashing the power of large-scale unlabeled data. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 10371-10381).

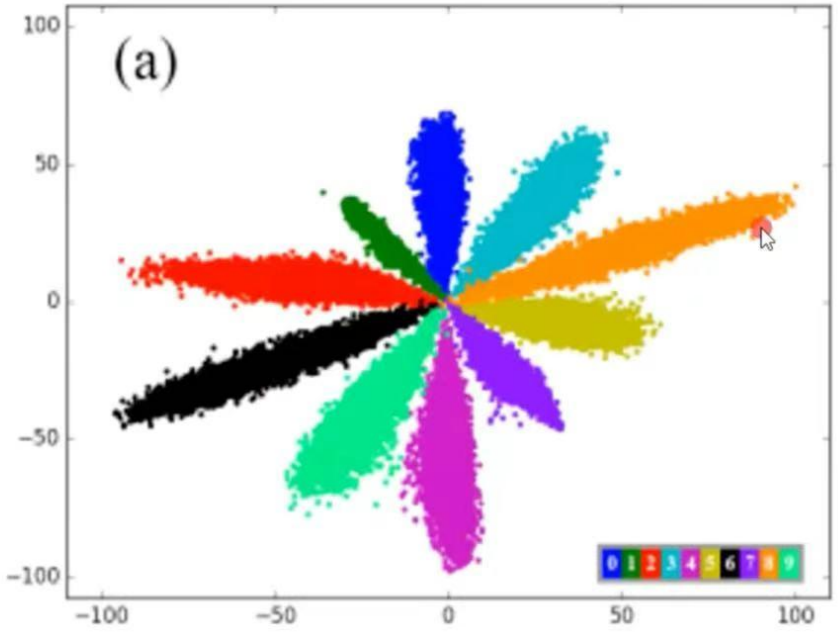
模型架构——stage2


Second Stage
fine-tuning
N Contrastive samples
X
Y
Z
Camera
Randn
R
P
Y
Encoder
(pretrained)
Feature Map
UV
Decoder
pred_uv
$\mathcal{L}_{MMD}$
$\mathcal{L}_{uv}$
$\mathcal{L}_{Con}$

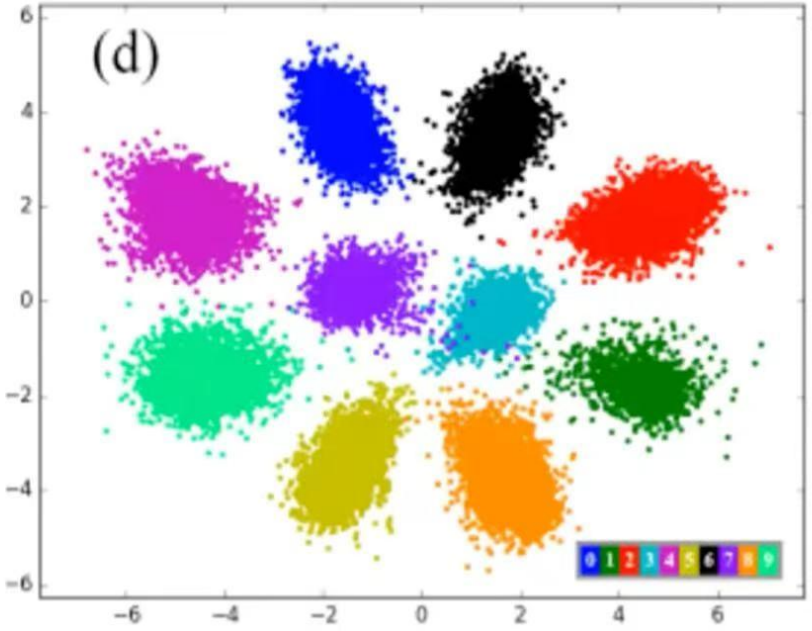

Second Stage
fine-tuning(Adapter)
N Contrastive samples
X
Y
Z
Camera
Randn
R
P
Y
Encoder
?
Adapter
Feature Map
UV
Decoder
pred_uv
$\mathcal{L}_{MMD}$
$\mathcal{L}_{uv}$
$\mathcal{L}_{Con}$

模型架构——约束方式

特征空间对比



表征学习



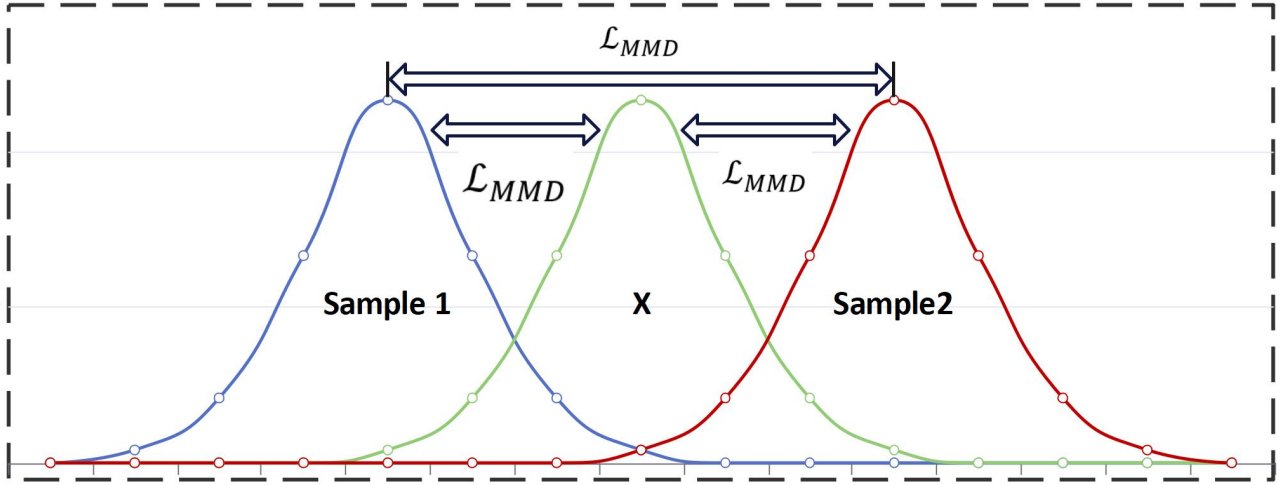
度量学习

模型架构——MMD_loss

✅ 1. MMD 定义公式：

MMD 的平方定义如下：

$$\mathrm{MMD}^2(\mathcal{D}_s, \mathcal{D}_t) = \left\| \frac{1}{n} \sum_{i=1}^{n} \phi(\mathbf{x}_i^s) - \frac{1}{m} \sum_{j=1}^{m} \phi(\mathbf{x}_j^t) \right\|_{\mathcal{H}}^2$$



✅ 2. 用核函数展开：

利用核技巧（kernel trick），该式可以展开为：

$$\mathrm{MMD}^2 = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{i'=1}^{n} k(\mathbf{x}_i^s, \mathbf{x}_{i'}^s) + \frac{1}{m^2} \sum_{j=1}^{m} \sum_{j'=1}^{m} k(\mathbf{x}_j^t, \mathbf{x}_{j'}^t) - \frac{2}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} k(\mathbf{x}_i^s, \mathbf{x}_j^t)$$

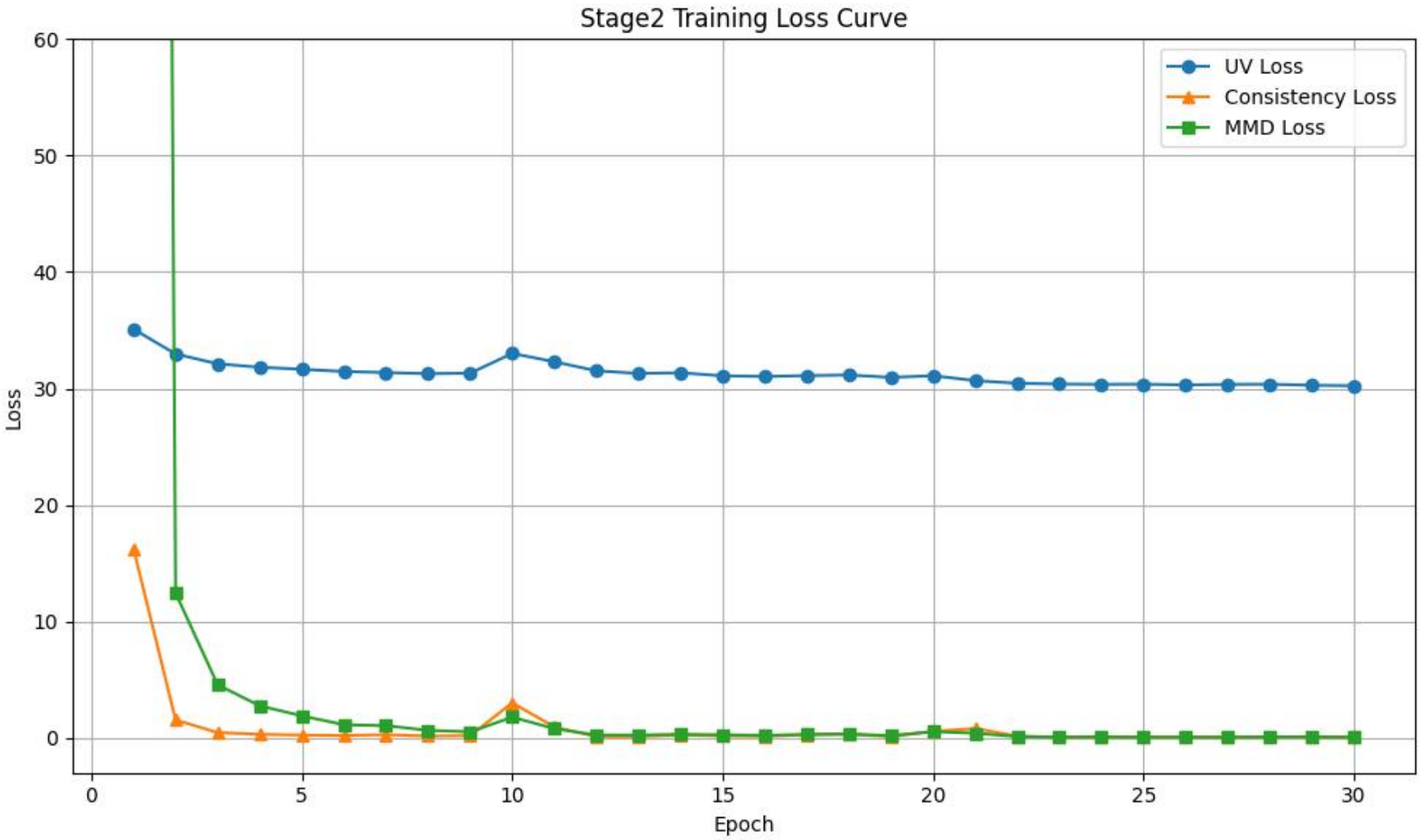
你代码中默认了 $n = m = \text{batch_size}$，所以公式可以简化为：

$$\mathrm{MMD}^2 = \frac{1}{n^2} \sum_{i,i'} k(\mathbf{x}_i^s, \mathbf{x}_{i'}^s) + \frac{1}{n^2} \sum_{j,j'} k(\mathbf{x}_j^t, \mathbf{x}_{j'}^t) - \frac{2}{n^2} \sum_{i,j} k(\mathbf{x}_i^s, \mathbf{x}_j^t)$$

第三部分

实验结果

实验结果-训练过程


Stage2 Training Loss Curve
UV Loss
Consistency Loss
MMD Loss
Loss
Epoch
60
50
40
30
20
10
0
0
5
10
15
20
25
30

实验结果-Code

Embodied-AI / final /

Add file

WMZ0518 final

3b57614 · 1 minute ago History

Name	Last commit message	Last commit date
..		
29.76.csv	final	1 minute ago
EU_stage1_31.14.pth	final	1 minute ago
EU_stage2_30.252.pth	final	1 minute ago
adapter.py	final	1 minute ago
adapter_test.py	final	1 minute ago
adapter_valid.py	final	1 minute ago
test.py	final	1 minute ago
train.py	final	1 minute ago
train_instruction	final	1 minute ago
train_mlp.py	final	1 minute ago

实验结果-Final Result

Leaderboard

Blind Projector

Showing results for: **Blind Projector**

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9	zhy-jty	31.5017
10	zjy_xyf	31.6046
11	gzr_qzh	31.7282
12	zqr_cqx	32.3640

Final Result : 30.7384 (1/12)

实验结果-Ablation Study

Method	Results
xyz世界坐标数据输入预测	58.5654
6维坐标数据输入预测	32.2131
仅第一阶段训练后预测	31.8732
第二阶段锁住Encoder融入Adapter	31.5673
第二阶段解锁Encoder (only $L_{u,v}$)	32.0962
第二阶段解锁Encoder ($L_{u,v}$ + L_{con})	31.2372
第二阶段解锁后融入L_{MMD} L_{Con} **(Best)**	**30.7384**

第四部分

讨论分析

分析总结

1. “用知识分析问题，用数据解决问题”——必须结合实际

2. 不能盲目追从“健壮先进”backbone——“设计为先，适用为王”

3. 学习方法和思想并懂得运用——“站在巨人的肩膀上看世界”

4. 锻炼“百折不挠”的科研精神——不要怀疑自己，向“SOTA妥协”

实验结果-Final Result

259	A: Blind Projector	2025-07-05 20:46:00	Completed	34.0138
251	A: Blind Projector	2025-07-05 20:28:53	Completed	31.8089
237	A: Blind Projector	2025-07-05 19:01:20	Completed	31.7238
231	A: Blind Projector	2025-07-05 18:35:08	Completed	32.3957
229	A: Blind Projector	2025-07-05 18:15:17	Completed	32.0928
194	A: Blind Projector	2025-07-05 16:56:35	Completed	32.6652

2025 / 7 / 5 / 23:59
Rank 8

Leaderboard

Blind Projector

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10	zjy_xyf	31.6046
11	gzr_qzh	31.7282
12	zqr_cqx	32.3640

2025 / 7 / 6 / 8:00
Rank 1 !!!

References

1. Piccinelli, L., Yang, Y. H., Sakaridis, C., Segu, M., Li, S., Van Gool, L., & Yu, F. (2024). UniDepth: Universal monocular metric depth estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 10106-10116).
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3. Yang, L., Kang, B., Huang, Z., Xu, X., Feng, J., & Zhao, H. (2024). Depth anything: Unleashing the power of large-scale unlabeled data. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 10371-10381).
4. Bai, Jinze, et al. "Qwen technical report." arXiv preprint arXiv:2309.16609 (2023).
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6. Wang, Wei, Yuan Sun, and Saman Halgamuge. "Improving MMD-GAN training with repulsive loss function." arXiv preprint arXiv:1812.09916 (2018).

谢谢各位老师的指导和SIGS提供的广阔平台

请您批评指正！

日期：6 / 7 / 2025